All of Us (AoU) Research Program Electronic Health Records



1. Acute myeloid leukemia phenotype

1. **Case definition**
   1. **At least one standard SNOMED condition or observation code corresponding to acute myeloid leukemia.**
   2. Passed genomic quality control.
2. **Control definition**
   1. **Lacks a standard SNOMED condition or observation code corresponding to acute myeloid leukemia.**
   2. Possesses electronic health records.
   3. Passed genomic quality control.

2. European-like ancestry sample

1. **N cases: 265**
2. **N controls: 169,706**

Gene-Based Tests

Gene	Cauchy p-value	Bonferroni p-value	cMAC
DNMT3A	4.44×10^{-10}	8.03×10^{-6}	2645
TET2	2.92×10^{-8}	5.27×10^{-4}	1563
SRSF2	1.53×10^{-7}	2.77×10^{-3}	255
IDH2	1.34×10^{-6}	2.43×10^{-2}	300

Clonal Haematopoiesis of Indeterminate Potential (CHIP)
Variant-Level QC Filters:

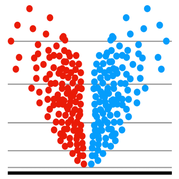
- Allele balance (AB): 0.35-0.65
- Read depth (DP): ≥20
- Exclusion of known CHIP-associated hotspot variants

Gene	AB, DP, and CHIP Hotspots Filtered		
	Cauchy p-value	Bonferroni p-value	cMAC
DNMT3A	1.57×10^{-5}	2.84×10^{-1}	2212
TET2	2.66×10^{-7}	4.81×10^{-3}	1524
SRSF2	1.00×10^{0}	1.00×10^{0}	46
IDH2	8.77×10^{-1}	1.00×10^{0}	243

Gene-Set Rare-Variant Burden Score

Gene-Set References

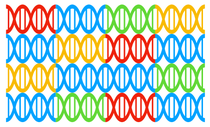
Bulk RNA-Seq



Single-Cell RNA-Seq



Somatic Mutation



Gene-Set Logistic Regression

$$\text{Phenotype} \sim \beta 1 \times \{\text{DNMT3A, TET2, SRSF2, IDH2}\} + \beta 2 \times \text{Age} + \beta 3 \times \text{Sex}$$

Cross-Reference Databases

NIH NATIONAL CANCER INSTITUTE Genomic Data Commons

ClinVar
Clinically relevant variation

Leave-One-Variant-Out Analysis

